METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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RELIEF LOAD BASIS

PSV-201, proposed reliefs, HV-205 depressuring · R-201 synthesis loop

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DOCUMENT TITLE : Relief Load Basis

PROJECT : Methanol Converter Ver 1 · OTS

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Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
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1 Purpose, basis and methods

This document lists every relief device on the EPC P&IDs MEOH-PID-001/002 (PSV-201 on V-201, set 88 bar g; HV-205 manual blowdown) and the reliefs that the HAZOP, the Safety layers study and this review find missing; for PSV-201 it evaluates the credible overpressure scenarios of API 521, gives the relieving rate of each, the governing case and the API 520 orifice. Rates and fluids come from the simulator model at its design point and from offline model runs; design pressures from the equipment list MEOH-V1-ME-EQL-001. Assumptions are marked.

Methods (typical practice). API 520 Part I critical gas sizing (SI): $A = W / (C K_d P_1 K_b K_c) \sqrt{(T Z / M)}$, A in mm², W in kg/h, P_1 in kPa a, T in K, $C = 0.03948 \sqrt{(k (2/(k+1))^{(k+1)/(k-1)})}$; K_d 0.975 (preliminary), K_b 1 (conventional valve, built-up back-pressure ≤ 10 %), K_c 1. Allowable overpressure 10 % (single valve), 21 % for fire. API 521 fire heat input to a wetted vessel $Q = 43\,200\ F\ A^{0.82}$ W (adequate drainage and fire fighting). Tube rupture: choked flow through the two open ends of one tube, C_d 0.6; relief is needed if the low-pressure side design is below 10/13 of the high-pressure side design (two-thirds rule in older editions). API 526 orifice letters. No credit for the SIS or for control action that helps; credit for outlets that stay open at their normal opening.

2 Relief devices

Device	Where	Set / size	Protects	Status
PSV-201	V-201 top → flare (MEOH-PID-002)	88 bar g; N orifice (4N6) selected here	Whole loop, zones A and B (V-201, E-203, E-202, E-201, E-204 tubes, R-201, C-201)	On the P&ID (sizing TBC in note 6 → this document)
HV-205	V-201 vapour → flare (MEOH-PID-002)	manual blowdown; restriction orifice ≈ 49 mm (this document)	Depressuring in a fire or leak (API 521 §5.20)	On the P&ID (RO TBC); remote operation recommended (HAZOP R15)
PSV-205	Let-down drum (downstream of LV-204, off-sheet)	10 bar g (assumed)	Let-down drum against gas blow-by	Proposed (HAZOP R03) — not on the P&ID
TSV-203 / E-203 CW relief	E-203 cooling-water shell	10 bar g (assumed)	CW side: thermal expansion and tube rupture	TSV-203 recommended in the Safety layers tab; tube-rupture relief proposed here
E-204 steam-side PSV	E-204 shell (HP steam)	45 bar g (assumed)	Steam side against tube rupture	Proposed here (tag TBC)

3 PSV-201 overpressure scenarios (set 88 bar g, P_1 97.81 bar a)

Front end: the model make-up source is 87 bar a, so in the model the loop cannot reach the set (purge-shut run 85.99 bar g). This basis assumes the front end delivers 402.2 kNm³/h (110 % of design) up to a settle-out pressure of about 101 bar a (assumed, EQ-er-01). Normal consumption by reaction and condensation 260.5 kNm³/h; purge at its normal opening at the relieving pressure 135.2 kNm³/h.

No.	Scenario (API 521)	Credible	Relief load basis	kNm³/h	W (kg/h)	M	T (°C)	k	Z	A (mm²)	Orifice
S1	Blocked outlet: purge PV-203 shut	Yes	PV-203 fails closed (air) or PIC-203 left in MAN at 0 %; C-201 and the reaction keep running. Make-up at the front-end maximum 402.2 kNm³/h (110 % of design, assumed) minus the normal consumption by reaction and condensation (365.6 − 105.1 = 260.5 kNm³/h). Relieved gas: loop gas.	141.7	36,847	5.83	40	1.354	1.03	1,076	K
S2	**C-201 stops with make-up on (no consumption)**	Yes	C-201 motor trip or local power failure: no circulation, so no reaction and no condensation. FIC-202 keeps FV-202 open; the front end delivers 402.2 kNm³/h (assumed). No credit for the SIS (FSLL-201 / I-2) — API 521 practice. Credit only for the purge at its normal opening at the relieving pressure (135.2 kNm³/h). Relieved gas taken as make-up gas (heavier than loop gas: conservative).	266.9	**115,986**	9.74	40	1.364	1.03	**2,613**	**N**
S3	Control-valve failure: FV-202 wide open	Yes	Flow through FV-202 is limited by the front end (402.2 kNm³/h); reaction, condensation and the purge at its normal opening continue: 402.2 − 260.5 − 135.2 = 6.4 kNm³/h. The model run (FV-202 100 %, PIC-203 in AUTO) levels at about 80.6 bar g with the bed trip at 392 s.	6.4	1,670	5.83	40	1.354	1.03	49	D
S4	External fire on V-201 (wetted, crude methanol)	Yes	API 521: $Q = 43\,200\ F\ A^{0.82}$ (adequate drainage and fire fighting, F = 1 bare vessel); wetted area to NLL 1.38 m above the bottom tangent, vessel bottom 3 m above grade (assumed): A = 37.4 m², Q = 841 kW. At the relieving pressure the crude is near its critical point, so the API 521 minimum latent heat 115 kJ/kg is used (assumed). Overpressure 21 %.	20.5	26,334	28.80	250	1.194	1	419	H
S5	Cooling-water failure to E-203	Yes	Crude methanol no longer condenses: 131.5 t/h stays in the vapour. No credit for the purge (conservative). Relieved as crude vapour at the E-202 hot outlet temperature 156 °C, Z = 1 (conservative). In the model (75 % CW loss, PIC-203 in AUTO) the loop peaks at 78.6 bar g.	102.4	131,516	28.80	156	1.194	1	2,083	M
S6	Tube rupture E-201 / E-202	No	Both sides are loop gas in zone B (design 100 bar g each): the low-pressure side is far above 10/13 of the high-pressure side — no relief case.	—	—	—	—	—	—	—	—
S7	Tube rupture E-203 / E-204 into the loop	No	The loop is the high-pressure side (cooling water ~4 bar g, steam 40 bar g): a rupture sends loop gas out of the loop — relief on the utility side, see the other devices.	—	—	—	—	—	—	—	—
S8	Abnormal heat input: HV-214 steam passing / open	No	Up to 30 MW into the bed-1 feed (steam valve passing); E-203 removes the heat and TIC-211 / TSHH-2xx act: temperature, not pressure, case.	—	—	—	—	—	—	—	—
S9	Runaway reaction	No	Methanol synthesis reduces the moles (4 → 2 per CH_3OH): a runaway lowers the pressure. Temperature protection TSHH-2xx (I-1) — see the SRS tab.	—	—	—	—	—	—	—	—
S10	Instrument-air failure	No	FV-202, PV-203 and LV-204 fail closed: no inflow, inventory held. The model stays at 75 bar g.	—	—	—	—	—	—	—	—
S11	Gas blow-by through LV-204 / LB-204	No (for PSV-201)	Overpressures the let-down drum, not the loop: PSV-205 (proposed, HAZOP R03), see the other devices.	—	—	—	—	—	—	—	—

Governing: S2 C-201 stops with make-up on (no consumption) — 115,986 kg/h → 2,613 mm² → API 526 N (2800 mm², typical body 4N6), rated 124,298 kg/h.

3.1 Worked calculation — governing case

Worked example — PSV-201, governing case S2 (C-201 stops with make-up on (no consumption)). Make-up at the front-end maximum 402.2 kNm³/h less the purge at its normal opening at relieving pressure 135.2 kNm³/h = **266.9 kNm³/h**; as make-up gas (M = 9.739 kg/kmol): W = 266.9 × 1000 / 22.414 × 9.739 = **115,986 kg/h**. Relieving pressure P_1 = 88 × 1.10 + 1.013 = 97.81 bar a = 9781 kPa; T = 313.2 K; Z = 1.03 (assumed); k = $c_p/(c_p - R)$ = 1.364. C = 0.03948 √(1.364 × $(2/2.364)^{6.49}$) = 0.02679. A = 115,986 / (0.02679 × 0.975 × 9781 × 1 × 1) × √(313.2 × 1.03 / 9.739) = **2,613 mm²** → API 526 **N** (2800 mm², 4N6), rated 124,298 kg/h. Inlet loss at rated flow through DN150 (Sch 160, ID 131.8 mm), 3 m equivalent: 0.94 bar = 1.07 % of set (≤ 3 %).

4 Other relief devices (not on the P&IDs)

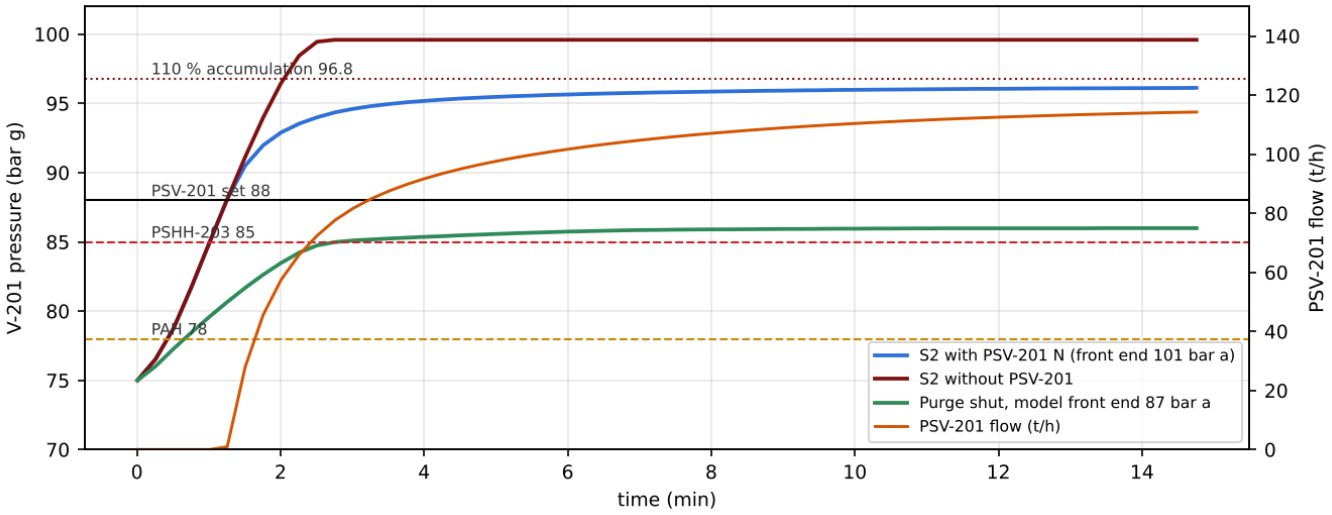
Device	Scenario	Rule / basis	W (kg/h)	Set, P_1	A (mm²)	Selected
PSV-205 (proposed, HAZOP R03)	Gas blow-by from V-201 through LV-204 and LB-204 wide open after the V-201 level is lost	C_v of LV-204 from the model (K_{lv} = 0.0108 m³/s per √bar at SG 0.814 → K_v 35.1, C_v 40.6); LB-204 adds 60 % (model); choked gas W = 27.3 C_v Y √(x_T P_1 ρ_1), x_T 0.7, Y 2/3 (assumed)	35,022	10 bar g (let-down drum design, assumed); P_1 12.01 bar a	8,201	R (6R8)
E-203 CW side (TSV-203 + tube-rupture relief, proposed)	Tube rupture: loop gas at up to 88 bar g into the cooling-water shell (design 10 bar g); TSV-203 for blocked-in cooling water (thermal)	10/13 rule: 10 < 10/13 × 100 = 76.9 bar g → relief required. Two open tube ends, ID 14.83 mm, C_d 0.6, HP side 88 bar g (PSV-201 set, assumed), effluent gas M 7.16 at 156 °C	6,342	10 bar g; P_1 12.01 bar a	1,575	L (4L6) or bursting disc; TSV-203 DN20 × DN25 (thermal, typical)
E-204 steam side (PSV, proposed; tag TBC)	Tube rupture: loop gas into the HP steam shell (design 45 bar g, assumed)	10/13 rule: 45 < 76.9 bar g → relief required. Same opening and C_d, loop gas M 6.57 at 254 °C (E-201 cold outlet)	5,509	45 bar g; P_1 50.51 bar a	375	H (2H3)

4.1 Fire case data (S4)

V-201: ID 4.2 m, liquid at NLL 1.38 m above the bottom tangent, vessel bottom 3 m above grade (assumed); wetted area to ≤ 7.6 m above grade 37.4 m²; Q = 43 200 × 1 × $37.4^{0.82}$ = 841 kW; latent heat 115 kJ/kg (API 521 minimum near the critical point, assumed) → W = 26,334 kg/h. Gas-filled equipment (R-201, the exchangers) cannot be protected by a PSV in a fire — the unwetted wall overheats first; their protection is depressuring (section 6), fire-proofing and layout.

5 Model relief runs

Offline copies of the simulator model (trips off, c.opt.noTrip) driven into the relief cases. The model has no PSV-201; the run emulates it after each step (lift linear from the set to +10 %, API 520 capacity of the chosen orifice with K_d 0.975 on the gas in V-201) and removes the relieved gas from the loop inventory. The front-end pressure (model K.Pmu) is raised only for the run and restored afterwards.

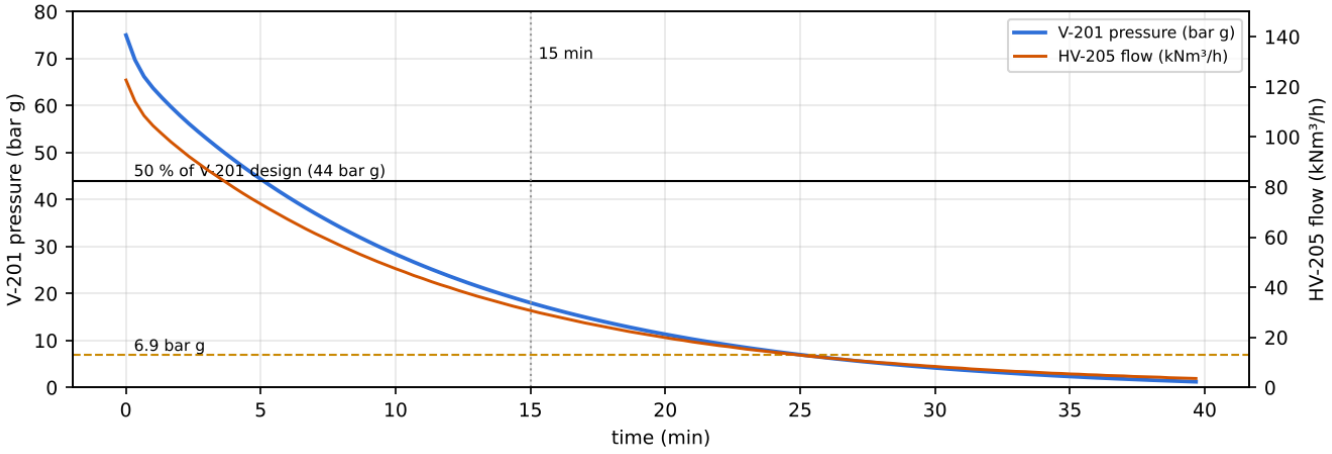


Run	Front end (bar a)	PSV-201	Peak V-201 (bar g)	Set reached (s)	Peak PSV flow (kg/h)
Purge shut, design make-up (Safety tab case)	87	none	85.99	—	—
S2 C-201 stops, make-up 110 % (governing)	101	N	96.12	76	114,368
S2 without PSV-201	101	none	99.59	76	—
S1 purge shut, make-up 110 %	101	N	90.95	124	30,182
S3 FV-202 wide open, PIC-203 AUTO	101	N	90.05	56	24,558
S2 with trips as installed	101	N	76.78	—	FSLL-201 circulation below 45 % of design at 17 s

With the selected orifice the governing case stays at 96.12 bar g, within 110 % of the set (96.8 bar g); the emulated peak flow 114,368 kg/h matches the S2 load 115,986 kg/h. With the SIS active FSLL-201 cuts the make-up first: the PSV is the last layer.

6 Depressuring through HV-205

ESD (I-5: C-201 stopped, XV-202 closed) and HV-205 fully open, model blowdown flow 0.02 × P_{abs} kmol/s. A restriction orifice of ≈ 49 mm (C_d 0.61, choked) passes the same 31,909 kg/h at 75 bar g. V-201 reaches 50 % of its design pressure (44 bar g) after 5.1 min (API 521: about 15 min for walls ≥ 25 mm — met) and 6.9 bar g after 25.1 min (older guideline, not met). The model is isothermal: the blowdown chilling and the minimum metal temperature need a dynamic study (EQ-em-05).



7 Checks at issue

Check	Value	Reference	Result
PSV-201 set ≤ lowest design pressure protected	88 bar g	EQL minimum 88 bar g	✓

Check	Value	Reference	Result
PSV-201 set > PSHH-203 trip	88 bar g	85 bar g	✓
PSHH-203 ≤ 95 % of set (spring simmer margin)	96.6 %	≤ 95 % (assumed practice)	✗ (EQ-er-02)
Normal operating ≤ 90 % of set	85.2 %	≤ 90 %	✓
Governing area ≤ selected orifice	2,613 mm²	N 2800 mm²	✓
Inlet loss ≤ 3 % of set	0.94 bar (1.07 %)	DN150 (Sch 160, ID 131.8 mm), 3 m equivalent (assumed)	✓
Model purge-shut peak below the set (model front end)	85.99 bar g	88 bar g; Safety tab 85.99	✓
Model governing case ≤ 110 % of set	96.12 bar g	96.8 bar g	✓
Depressuring to 50 % design within 15 min	5.1 min	≤ 15 min	✓
Let-down drum blow-by relief on the P&ID	not drawn	HAZOP R03	✗ (EQ-er-05)
E-203 CW-side tube-rupture relief on the P&ID	not drawn	10/13 rule	✗ (EQ-er-03)
E-204 steam-side tube-rupture relief on the P&ID	not drawn	10/13 rule	✗ (EQ-er-04)

The Relief & vent tab recomputes these checks from the live model, its alarm list and trip constants, and the equipment list each time it is opened.

8 Engineering queries

No.	From → to	Issue	Proposed
EQ-er-01	MEOH-V1-PE-RLB-001 → MEOH-V1-PE-HMB-001 (battery limits) / front-end	The front end's maximum delivery pressure is not defined. In the model the make-up source is 87 bar a (≈ 86 bar g), so the loop can never reach the PSV-201 set (purge-shut run: 85.99 bar g). The relief loads S1–S3 assume a front end able to deliver 402 kNm³/h (110 % of design) at the relieving pressure, i.e. a shut-in / settle-out pressure of about 100 bar g (assumed).	Confirm the syngas compressor shut-in (settle-out) pressure and maximum capacity. If it cannot exceed 88 bar g, S1–S3 are not credible and the fire case S4 governs (smaller PSV); if it is higher, keep the governing case of this document.
EQ-er-02	MEOH-V1-PE-RLB-001 → MEOH-V1-IN-CE-001 / MEOH-V1-IN-SRS-001	PSHH-203 trips at 85 bar g = 96.6 % of the PSV-201 set. A conventional spring valve can simmer and leak from about 93–95 % of its set, i.e. before or at the trip; the operating margin between PAH 78, the trip and the set is only 3 bar.	Use a pilot-operated PSV-201 (tight to ~98 % of set), or lower PSHH-203 to ≤ 83 bar g (check spurious trips against PAH 78 bar g), or raise the zone A design / set pressure.
EQ-er-03	MEOH-V1-PE-RLB-001 → MEOH-PID-002 / Safety layers tab (TSV-203)	E-203 cooling-water side (10 bar g, assumed) is below 10/13 of the process side (100 bar g): a tube rupture needs relief on the CW side, 6342 kg/h of gas → 1575 mm² (L). Neither TSV-203 (thermal, Safety layers tab) nor any CW-side relief is drawn on MEOH-PID-002.	Add TSV-203 and a tube-rupture relief (PSV L or bursting disc to a safe location / flare) on the E-203 shell, or raise the CW-side design to ≥ 77 bar g; consider a dynamic tube-rupture check (liquid-full shell).
EQ-er-04	MEOH-V1-PE-RLB-001 → MEOH-PID-001	E-204 steam side (45 bar g, assumed) is below 10/13 of the process side: tube rupture sends 5509 kg/h of loop gas into the HP steam system; no steam-side relief is drawn.	Add a steam-side PSV on E-204 (H, set 45 bar g, to flare — it would relieve hydrogen-rich gas) or confirm the HP steam header relief covers it; keep HV-214 and the steam block valve closed outside start-up.
EQ-er-05	MEOH-V1-PE-RLB-001 → MEOH-PID-002 / HAZOP R03	HAZOP R03 asks for PSV-205 on the let-down drum sized for full gas blow-by; it is not on the P&ID. With LV-204 and LB-204 wide open the blow-by is about 35022 kg/h of loop gas; at a 10 bar g drum (assumed) this needs 8201 mm² (R).	Show PSV-205 (R) on the let-down drum; also apply HAZOP R02 (LSLL-204 closing a new XV-204) to reduce the demand.
EQ-er-06	MEOH-V1-PE-RLB-001 → MEOH-PID-002 note 6 / HAZOP R15	HV-205 restriction orifice is "TBC" on the P&ID. A 49 mm bore (C_d 0.61) reproduces the model blowdown rate (122.7 kNm³/h, 31909 kg/h at 75 bar g). With it the model reaches 50 % of the V-201 design in 5.1 min but 6.9 bar g only after 25.1 min, and HV-205 is a manual field valve.	Write the RO bore on the P&ID; make HV-205 a remotely operated, fail-open blowdown valve (HAZOP R15); confirm the flare capacity for the larger of PSV-201 and blowdown.
EQ-er-07	MEOH-V1-PE-RLB-001 → MEOH-PID-002	PSV-201 is a single valve with no isolation valves shown; it protects the whole loop, so it cannot be tested or replaced on line.	Install 2 × 100 % PSVs with interlocked (car-sealed open) inlet and outlet isolation valves, typical practice, and show them on the P&ID.

9 Assumptions

Front end: maximum delivery 110 % of design make-up up to ≈ 101 bar a settle-out (model boundary 87 bar a). Z = 1.03 for loop and make-up gas, 1.0 for crude vapour; k from the model heat capacities. Fire: F = 1, vessel bottom 3 m above grade, latent heat 115 kJ/kg. Tube rupture: tubes 19.05 × 2.11 mm, C_d 0.6, high-pressure side at 88 bar g. Let-down drum design 10 bar g; E-203 CW side 10 bar g; E-204 steam side 45 bar g. LV-204 gas capacity from the model liquid coefficient, x_T 0.7, LB-204 = 60 % of LV-204 (model). PSV-201 inlet DN150 Sch 160, 3 m equivalent, f 0.015. HV-205 orifice C_d 0.61.